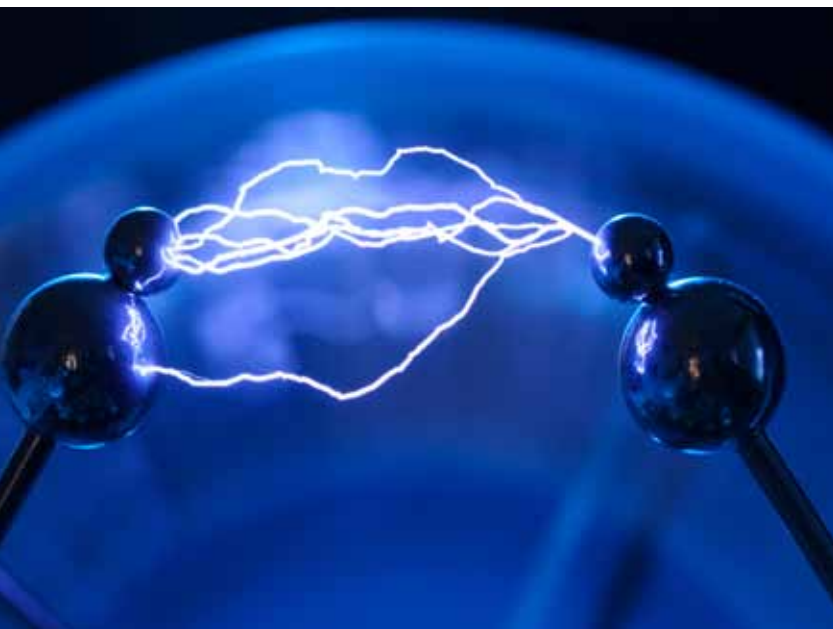


WHY ‘MODERN ICs FAIL ‘SILENTLY’ And Solution To The Problem

When failure margins shrink to a few volts, ioniser performance reveals why ESD risks go unnoticed and how they can be addressed.



AKANKSHA SONDHI
GAUR

Unexpected electrostatic discharge (ESD) losses often hide in plain sight. Production lines pass audits, operators follow every protocol, ionisers hum reassuringly in the background, yet devices still fail quietly, leaving engineers chasing ghosts inside an otherwise compliant setup. For many factories, the widening gap between visible control and invisible damage is driven by a single factor: today's electronics are far more sensitive than the standards designed to protect them.

Across global manufacturing environments, ESD specialists are now drawing attention to two ionisation parameters that were once treated as minor details: decay time and offset voltage. As device architectures shrink, these parameters have moved from the margins to the centre of unexplained yield loss.

Why ioniser performance matters more than ever

For years, ionisers were treated as install-and-forget equipment. However, ionisers can become silent liabilities when key parameters drift or go unmeasured. Two metrics determine whether an ioniser is genuinely protecting sensitive electronics:

- **Decay time.** The speed at which a +1000V or -1000V charge is neutralised to within $\pm 100V$.

- **Offset voltage (ion balance).** The residual voltage left after neutralisation, essentially the ioniser's own bias.

In environments dominated by ultra-dense semiconductors, compliance with these limits no longer guarantees safety. Industry data illustrates how dramatically device sensitivity has shifted. Charged-device model (CDM) tolerances have collapsed for many advanced components.

Numerous 5G/RF (radio frequency), power management integrated circuit (PMIC), complimentary metal-oxide semiconductor (CMOS), and sensor SoCs now fail in the 30-50V range—levels that would have been considered ultra-sensitive only a few years ago. For AI accelerators and high-bandwidth memory controllers, real-world CDM sensitivities drop further, often below 30V. Recent 2024 findings from the EOS/ESD Association highlight an even harsher reality: some next-generation, high-density ICs exhibit sub-10V susceptibility.

When a device fails at 30V, an ioniser balanced at $\pm 35V$, while technically compliant, is already operating outside the device's tolerance. Standards remain baseline guidelines, not guarantees of protection.

Where hidden failures truly originate

In many factories, slow neutralisation or balance drift is the missing link behind